



Hall Ticket No. -

Code: S1235315

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME/AI&DS/IT) III Semester (Backlog) Examinations-May-June-2023

OOPS Using Java

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- a. Define Class and Object. [2M CO1 BTL1]
b. What is abstraction in java? [2M CO2 BTL1]
c. What are the different ways of thread creation? [2M CO3 BTL1]
d. What are exception handling keywords? [2M CO3 BTL1]
e. Define Serialization in Java. [2M CO4 BTL1]
f. Define event source and event listener. [2M CO5 BTL1]

Part-B

4X12=48M

- a) What is Scope and life time of a variable? Explain with an example. [6M CO1 BTL4]
b) Write a program to demonstrate Class, Object and Method. [6M CO1 BTL3]
- a) Difference abstract classes and interfaces. [4M CO2 BTL2]
b) Write a program to demonstrate multilevel inheritance. [8M CO2 BTL4]
- a) Explain exception handling in Java with an example. [6M CO3 BTL2]
b) Write a program to demonstrate thread creation using runnable interface. [6M CO3 BTL3]
- a) Explain the collection class hierarchy. [7M CO4 BTL3]
b) Demonstrate Scanner class with an example program. [5M CO4 BTL3]
- a) Explain the Delegation Event Model in AWT. [6M CO5 BTL2]
b) What are the different types of event listeners supported by Java? Explain. [6M CO5 BTL2]
- a) Demonstrate constructor overloading concept with the help of a program. [7M CO1 BTL2]
b) What is the use of adapter classes? Explain. [5M CO5 BTL2]



Hall Ticket No. -

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Stanley College of Engineering & Technology for Women (A)

B.E (AI&DS/IT) III Semester (Backlog) Examinations – May-June-2023

Database Management Systems

Time: 03Hours

Max. Marks-60

- Note: i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. What are the problems with File system and what is the need of DBMS? [2M CO1 L2]
b. List SQL Relational Set Operators with suitable examples. [2M CO2 L1]
c. What are the basic constructs of ER model? [2M CO3 L2]
d. What is difference between primary and secondary index? [2M CO4 L2]
e. Why backup and recovery is needed? [2M CO5 L3]
f. Distinguish between simple and composite attributes. [2M CO1 L2]

Part-B

4X12=48M

2. a) What is data redundancy? Explain Database Architecture and the functionality of each subsystem. [7M CO1 L2]
b) List the types of user and summarize any 6 responsibilities of DBA. [5M CO1 L2]
3. a) Explain Set and Joint operation in SQL. [6M CO2 L3]
b) List fundamental relational algebra operations and explain with example. [6M CO2 L3]
4. a) Illustrate 1NF and 2NF with suitable examples. [6M CO3 L2]
b) Illustrate the concept of generalization, specialization and aggregation with example. [6M CO3 L3]
5. a) Differentiate hashing techniques with an example. [4M CO4 L2]
b) What is serializability? Explain the concept of serializability with an example. [8M CO4 L2]
6. a) What is Concurrency Control? Explain Two phase locking protocol. [7M CO5 L2]
b) With a suitable example explain briefly the log base recovery with concurrent transaction. [5M CO5 L3]
7. a) Define ACID properties. Draw the transaction state diagram and explain states. [6M CO4 L2]
b) Illustrate any four integrity constraints with examples. [6M CO2 L3]



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Stanley College of Engineering & Technology for Women (A)

B.E (IT) III Semester (Backlog) Examinations-May-June-2023

Digital Electronics & Logic Design

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- a. State De Morgan's theorem. [2M CO1 L1]
- b. Construct 4X1 multiplexer. [2M CO2 L1]
- c. Describe an example of 2-input lookup table. [2M CO3 L1]
- d. Recite transition table of JK Flip-Flop. [2M CO4 L1]
- e. Recall basic design steps of a Mealy machine. [2M CO5 L1]
- f. Draw F.A using H.A circuit. [2M CO2 L3]

Part-B

4X12=48M

- a) Minimize the following Boolean expression using K-map
 $f = \sum m(0,1,2,3,5,6,7,8,9,10,13,14,15)$. [6M CO1 L3]
- b) Implement the function $F = X'Y + XY'$ using
(a) only NAND gates and (b) only NOR gates. [6M CO1 L3]
3. a) Design binary to gray code converter and implement with NOR gates. [6M CO2 L2]
- b) Design and explain 3 x 8 Decoder with neat logical circuit diagram. [6M CO2 L2]
4. a) Break down implementation of Boolean functions $F1 = XY'Z' + X'Y'Z + Y'Z$ and $F2 = XYZ + Y'Z$ by using Programmable Logic Array device. [6M CO3 L4]
- b) Outline the functioning of optimized 3 bit odd parity generator using PAL. [6M CO3 L4]
5. a) Explain the significance of excitation table with an example. Differentiate between combinational and sequential circuits. [6M CO4 L5]
- b) Design MOD 6 synchronous counter using T flip-flops. [6M CO4 L5]
6. a) Demonstrate Moore machine with example. Design a mealy machine to find 2's complement of a number. [6M CO5 L6]
- b) Demonstrate state minimization with an example. Design a non-overlapping sequence detector to detect an input sequence of '101' [6M CO5 L6]
7. a) Design SISO, PISO, Universal shift register. [6M CO4 L2]
- b) Explain full adder and full subtractor with a neat diagram. [6M CO1 L2]



Hall Ticket No. -

Code: S1235311

Stanley College of Engineering & Technology for Women (A)

B.E (IT) III Semester (Backlog) Examinations-May-June-2023

Discrete Mathematics

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- 1 a. What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$? [2M CO1 BTL2]
- b. Explain Euclidean Algorithm. [2M CO2 BTL3]
- c. Explain basics of counting. [2M CO3 BTL3]
- d. Use a truth table to verify the first De Morgan law $\neg(p \wedge q) \equiv \neg p \vee \neg q$ [2M CO4 BTL3]
- e. What is the expansion of $(x + y)^4$? [2M CO2 BTL3]
- f. Define degree of vertex. How many edges are there in a graph with 10 vertices each of degree six? [2M CO5 BTL2]

Part-B

4X12=48M

- 2 a. Define Relation. Explain the properties of binary relations with examples. [6M CO1 BTL2]
- b. Explain Power Set theorem. [6M CO1 BTL3]
- 3 a. Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? [6M CO2 BTL4]
- b. Explain the Laws of Logic. [6M CO5 BTL2]
- 4 a. Write a short note on Pigeon Hole Principles and its applications. [6M CO2 BTL2]
- b. In how many ways can 7 women and 3 men be arranged in a row if the 3 men must always stand next to each other? [6M CO2 BTL4]
- 5 a. Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology. [6M CO3 BTL5]
- b. State and explain which rule of inference is used in the argument:
If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow.
Therefore, if it rains today, then we will have a barbecue tomorrow. [6M CO3 BTL4]
- 6 a. Define group, monoids, semi groups and subgroups with examples. [6M CO4 BTL2]
- b. Explain Boolean Ring. [6M CO4 BTL5]
- 7 a. When it can be said that two graphs G_1 and G_2 are isomorphic. [6M CO5 BTL3]
- b. Explain shortest path problems. [6M CO5 BTL2]



Hall Ticket No. -

Code: S123532

Stanley College of Engineering & Technology for Women (A)**B.E (IT) III Semester (Backlog) Examinations – May-June-2023****Probability and Statistics****Time: 03Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a) For a frequency distribution of a variable x , the mean = 32, median = 30. Check whether it is a positively or negatively skewed. Justify your answer? [2 M CO1_BTL2]
 b) A random variable X has the following probability distribution.

X	1	2	3	4	5	6	7	8
$P(x)$	k	$2k$	$3k$	$4k$	$5k$	$6k$	$7k$	$8k$

Find the value of (i) k (ii) $P(x \leq 2)$

[2M CO 2_BTL2]

- c) Fit the straight line to the following data.

[2M CO_3 BTL2]

x	0	5	10	15	20
y	7	11	16	20	26

- d) If $y' = y - x$ and $y(0) = 2, h = 0.2$, then compute the value of y_1 by Euler's method.

[2M CO 4 BTL3]

- e) Compute $y(0.1)$ using Taylor's series method for $y' = x^2 y - 1, y(0) = 1$ given

$$y_0' = -1, y_0'' = 0, y_0''' = 2, y_0^{(4)} = -6 \text{ and } h = 0.1$$

[2M CO4_BTL3]

- f) Determine whether $v_1 = (1, 1, 1, 1), v_2 = (1, 2, 3, 2), v_3 = (2, 5, 6, 4), v_4 = (2, 6, 8, 5)$ are linearly independent or dependent?

[2M CO5_BTL2]

Part-B**4X12=48M**

2. a) Three machines A, B and C produce respectively 60%, 30% and 10% of the total number of items of a factory. The percentages of defective outputs of these machines are 2%, 3% and 4%. An item is selected at random and is found to be defective. Find the probability that the item was produced by machine C? [7MCO1_BTL 3]
 b) X is a poisson variate such that $p(X=2) = 9 p(X=4) + 90 P(X=6)$ find the mean and variance of X . [5M CO2_BTL2]

3. a) In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and S.D of the distribution [6M CO2 BTL4]
- b) A continuous random variable has p.d.f [6M CO2 BTL2]
- $$f(x) = k x e^{-\lambda x} \text{ for } x \geq 0, \lambda \geq 0 = 0 \text{ otherwise}$$
- Find i) k ii) Mean iii) Variance

4. a) Following are the ranks obtained by 10 students in two subjects. Statistics and Mathematics. To what extent the knowledge of the students in two subjects is related?

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	2	4	1	5	3	9	7	10	6	8

[6M CO3 BTL3]

- b) A pair of dice is thrown 360 times and the frequency of each sum is indicated below:

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	8	24	35	37	44	65	51	42	26	14	14

Would you say that the dice are fair on the basis of the chi-square test at 0.05 level of significance?

[6M CO3 BTL4]

5. a) The following data relate to the marks of 10 students in internal and university test for the maximum of 50.

Internal	25	28	30	32	35	36	38	39	42	45
University	20	26	29	30	25	18	26	35	35	46

Find out the regression line Y on X and determine the most likely internal mark for university marks 25

[8M CO3 BTL4]

- b) Find an approximate value of y when $x = 0.1$, if $\frac{dy}{dx} = x - y^2$ and $y = 1$ at $x = 0$ using

Taylor's method.

[4M CO4 BTL4]

- 6 a) Compute $y(0.4)$ by Milne's Predictor-Corrector method for the DE $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$.

[6M CO4 BTL3]

- b) Apply Runge-Kutta method of fourth order to find an approximate value of y when $x = 0.2$ given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$

[6M CO4 BTL4]

7. a) Show that the vectors $(2,2,0)$ $(3,0,2)$ $(2,-2,2)$ are linearly independent? [6M CO5 BTL3]

- b) Find SVD of the following Matrix

[6M CO5 BTL3]

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$
